



Assignment # 3

Subject: Database Systems -CS2005
Total Marks: 20

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Due Date: 9/11/2025

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Instructions to be strictly followed.

- For all questions involving SQL Queries:
 - o **Submit the SQL Scripts in a .txt file.**
- It should be obvious that submitting your work after the due date will result in zero points being awarded.
- Plagiarism (copying/cheating) and late submissions result in a zero mark.

Question#1 In 2025, the rise of **AI-powered cyberattacks**, **data breaches**, and **geopolitical cyber conflicts** has led the **Global Cyber Defense Alliance (GCDA)** to build a **centralized Cyber Threat Intelligence (CTI) database**. The GCDA database includes the following relations:

Relation	Attributes	Description
COUNTRY	(Country_ID, Country_Name, Region, GDP_Rank)	Stores country details
THREAT	(Threat_ID, Threat_Name, Category, Severity_Level)	Stores types of cyber threats (e.g., Ransomware, Phishing, AI Deepfake Attacks)
INCIDENT	(Incident_ID, Threat_ID, Country_ID, Date_Reported, Impact_Score)	Each reported cyber incident and its severity
RESPONSE_TEAM	(Team_ID, Country_ID, Team_Name, Specialization)	Cyber response teams in each country
ACTION_TAKEN	(Incident_ID, Team_ID, Action_Type, Resolution_Status)	Which team responded to which incident and what was done

You are a **Database Analyst** working for GCDA. Write the **Relational Algebra** expression and one line explanation of each command for the following:

Questions

1. List the names of countries that have reported critical-level threats (Severity_Level = "Critical") in 2025.

```

$$\pi \text{ Country\_Name } ($$
  

$$(\sigma \text{ Severity\_Level} = \text{'Critical'} \wedge \text{Date\_Reported LIKE '2025\%'})$$
  

$$(\text{THREAT} \bowtie \text{THREAT.Threat\_ID} = \text{INCIDENT.Threat\_ID}$$
  

$$\bowtie \text{INCIDENT.Country\_ID} = \text{COUNTRY.Country\_ID})$$
  

$$)$$

```

Explanation: Selects all countries that reported *Critical-level threats* in the year *2025*.

2. Retrieve the names of response teams that successfully resolved ransomware-related incidents.

```

$$\pi \text{ Team\_Name } ($$
  

$$(\sigma \text{ Category} = \text{'Ransomware'} \wedge \text{Resolution\_Status} = \text{'Resolved'})$$
  

$$(\text{THREAT} \bowtie \text{THREAT.Threat\_ID} = \text{INCIDENT.Threat\_ID}$$
  

$$\bowtie \text{INCIDENT.Incident\_ID} = \text{ACTION\_TAKEN.Incident\_ID}$$
  

$$\bowtie \text{ACTION\_TAKEN.Team\_ID} = \text{RESPONSE\_TEAM.Team\_ID})$$
  

$$)$$

```

Explanation: Finds the names of response teams that handled *ransomware* threats and marked them as *resolved*.

3. Find all countries that have not yet formed any cyber response team.

```

$$\pi \text{ Country\_Name } (\text{COUNTRY})$$

```

—

```

π Country_Name (

    COUNTRY ⋈ COUNTRY.Country_ID =
        RESPONSE_TEAM.Country_ID

)

```

Explanation: Shows countries listed in **COUNTRY** that do not appear in the **RESPONSE_TEAM** relation — meaning they have *no cyber response team* yet.

4. Display the names of threats that have been reported by more than one country.

```

π Threat_Name (

    σ count(Country_ID) > 1

    (γ Threat_ID, count(Country_ID)→NumCountries (INCIDENT ⋈
        THREAT))

)

```

Explanation: Groups incidents by **Threat_ID** and selects those appearing in *more than one country*.

5. List all AI-based attack incidents (Category = "AI Attack") along with their country names and impact scores greater than 80.

```

π Threat_Name, Country_Name, Impact_Score (
    (σ Category = 'AI Attack' ∧ Impact_Score > 80)
    (THREAT ⋈ THREAT.Threat_ID = INCIDENT.Threat_ID
        ⋈ INCIDENT.Country_ID = COUNTRY.Country_ID)
)

```

Explanation: Retrieves *AI Attack* incidents with an *impact score above 80* and displays related threat, country, and impact information.

Question#2 Consider a database with the following schema:

Cafe (cafeID, cafeName, city, managerID)
 Employee (empID, name, address, gender, salary, cafeID)
 Customer (custID, name, address, phone)
 Orders (orderID, custID, cafeID, totalAmount, orderDate)
 MenuItem (itemID, itemName, category, price, isAvailable)
 OrderDetails (orderID, itemID, quantity)
 Supplier (supplierID, name, city, contactNo)
 Supplies (supplierID, itemID, supplyDate, cost)

Describe the relations that would be produced by the following relational algebra operations (Textual meaning required) and also provide sql queries of each algebraic expression:

1. $\pi_{\text{itemName}} (\sigma_{\text{isAvailable} = 0 \wedge \text{price} > 500} (\text{MenuItem}))$

List the names of menu items that are **not available** and **cost more than 500**.

```
SELECT itemName
FROM MenuItem
WHERE isAvailable = 0 AND price > 500;
```

2. $\pi_{\text{itemName}} (\sigma_{(\text{category} = \text{'Beverage'} \wedge \text{price} > 300) \vee \text{category} = \text{'Dessert'}} (\text{MenuItem}))$

List the item names that are **beverages priced above 300** or **belong to the dessert category**.

```
SELECT itemName
FROM MenuItem
WHERE (category = 'Beverage' AND price > 300)
OR category = 'Dessert';
```

3. $\pi_{\text{itemName}, \text{price}} (\sigma_{\text{category} = \text{'Snack'} \wedge \text{isAvailable} = 1} (\text{MenuItem}))$

Retrieve the **name and price** of all **available snack items**.

```
SELECT itemName, price
FROM MenuItem
WHERE category = 'Snack' AND isAvailable = 1;
```

4. $\pi_{\text{itemName}, \text{price}/100} (\sigma_{\text{category} = \text{'Snack'} \wedge \text{isAvailable} = 1} (\text{MenuItem}))$

List snack items with their prices **converted to hundreds (price/100)** to show approximate cost.

```
SELECT itemName, price / 100 AS price_in_hundreds
FROM MenuItem
WHERE category = 'Snack' AND isAvailable = 1;
```

5. $\pi_{\text{supplierID}} (\text{Supplies} \bowtie \text{MenuItem} \bowtie \sigma_{\text{itemName} = \text{'Cappuccino'}} (\text{MenuItem}))$

Find the **supplier IDs** who supply the item **'Cappuccino'**

```
SELECT DISTINCT s.supplierID
FROM Supplies s
JOIN MenuItem m ON s.itemID = m.itemID
WHERE m.itemName = 'Cappuccino';
```

6. $\pi_{\text{Employee.name}, \text{Cafe.cafeName}} (\sigma_{\text{Employee.cafeID} = \text{Cafe.cafeID} \wedge \text{Employee.address} = \text{Cafe.city}} (\text{Employee} \bowtie \text{Cafe}))$

List the names of employees who **live in the same city as the cafe they work in**, along with the cafe name.

```
SELECT e.name AS EmployeeName, c.cafeName
```

```
FROM Employee e
JOIN Cafe c ON e.cafeID = c.cafeID
WHERE e.address = c.city;
```

7. $\pi_{\text{empID}} (\sigma_{\text{salary} \neq 50000} (\text{Employee}))$

Find employee IDs of all employees **whose salary is not equal to 50,000**.

```
SELECT empID
FROM Employee
WHERE salary <> 50000;
```

8. $\pi_{\text{supplierID}} (\sigma_{\text{city} = \text{'Karachi'}} (\text{Supplier})) \cap \pi_{\text{supplierID}} (\sigma_{\text{city} = \text{'Lahore'}} (\text{Supplier}))$

Find supplier IDs of suppliers who have **branches in both Karachi and Lahore**.

```
SELECT supplierID
FROM Supplier
WHERE city = 'Karachi'
INTERSECT
SELECT supplierID
FROM Supplier
WHERE city = 'Lahore';
```

9. $\pi_{\text{itemName}} (\text{MenuItem}) - \pi_{\text{itemName}} (\text{MenuItem} \bowtie \text{Supplies})$

List the item names that are **not supplied by any supplier** (e.g., house-made items).

```
SELECT itemName
FROM MenuItem
WHERE itemID NOT IN (SELECT itemID FROM Supplies);
```